A Neural Signature of Neuroticism?
Exploring the Functional Connectome to Predict Personality

Jana Bormann, Micha Burkhardt, Andrea Hildebrandt
Carl von Ossietzky Universität Oldenburg, Psychological Methods and Statistics

Carl von Ossietzky Universität Oldenburg

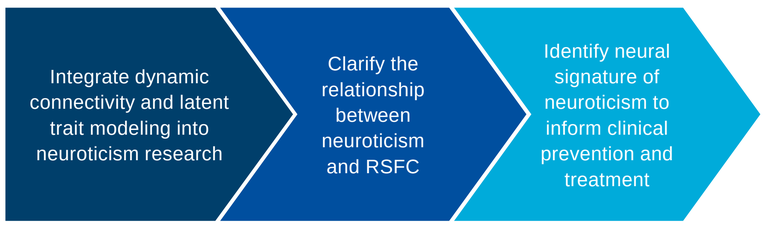
I. Introduction

Background / Rationale

Neuroticism:
A personality trait that serves as a **general vulnerability factor**, predicting a wide range of **mental disorders** (e.g., disorders of anxiety, mood, eating, substance use, and personality) as well as **physical health problems** (e.g., cardiovascular, respiratory, and gastrointestinal conditions) [1]

Functional Connectivity:
Studies investigating the **neural correlates** of neuroticism indicate that resting-state functional connectivity (RSFC) can **successfully predict** individual differences in the trait [2,3]. However, findings remain **inconsistent** [4], most studies rely on **static** connectivity approaches, and neuroticism is rarely modeled as a **latent construct.**

Aim and Relevance of This Study



Research Questions

Does **a latent variable model** adequately capture the structure of **neuroticism** in the Human Connectome Project (HCP) young adult dataset?

To what extent do global **static** and **dynamic** measures of RSFC **predict latent neuroticism traits** in healthy young adults?

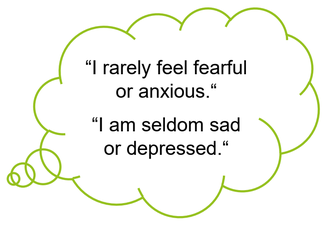
II. Methods

Measurement Model

Data and Instrument:
- **Sample**: *N* = 1198 healthy young adults (HCP-YA, 2025 data release) with a mean age of 28.84 years (range: 22-37)
- **Scale**: NEO-Five Factor Inventory (NEO-FFI) Neuroticism subscale (12 items; see thought bubble for two (reverse-keyed) example items).

Measurement Model Selection:
- **Estimation**: Utilized item response theory (IRT) via the *mirt* package to identify optimal latent structure
- **Comparison**: Evaluated four competing models
- **Selection Criteria**: χ^2 Likelihood Ratio Tests, AIC/BIC, and factor loading magnitudes

Model Approximation:
The optimal IRT structure was approximated via Confirmatory Factor Analysis (CFA) using the *lavaan* package



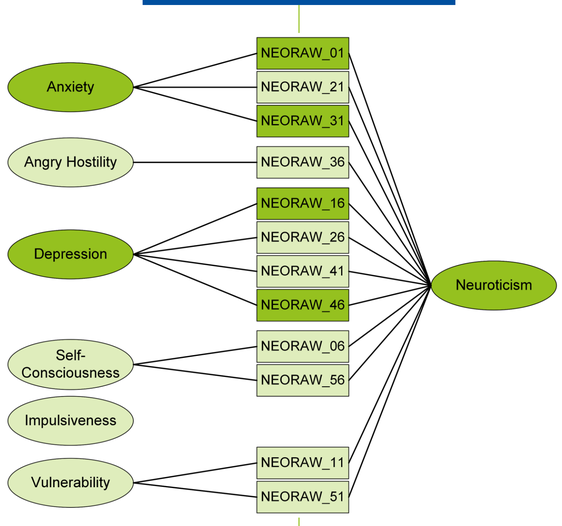


Figure 1: Allocation of items to specific neuroticism facets (left) and the general neuroticism factor (right). Non-transparent latent factors represent the components retained in the final bifactor model, while non-transparent items indicate those used to identify the specific facets. Notably, the general neuroticism factor was identified using all 12 items.

III. Results

Selected Model:
- **Bifactor Structure**: General neuroticism factor (g) and two specific facets – anxiety (s_anx) and depression (s_dep) – each comprising their two strongest items
- **Superior Fit**: Outperformed unidimensional and more complex models

Latent Structure Properties:
- **Factor loadings**: All standardized loadings > 0.3
- **Model Fit**: Statistics were excellent across both frameworks

approximates

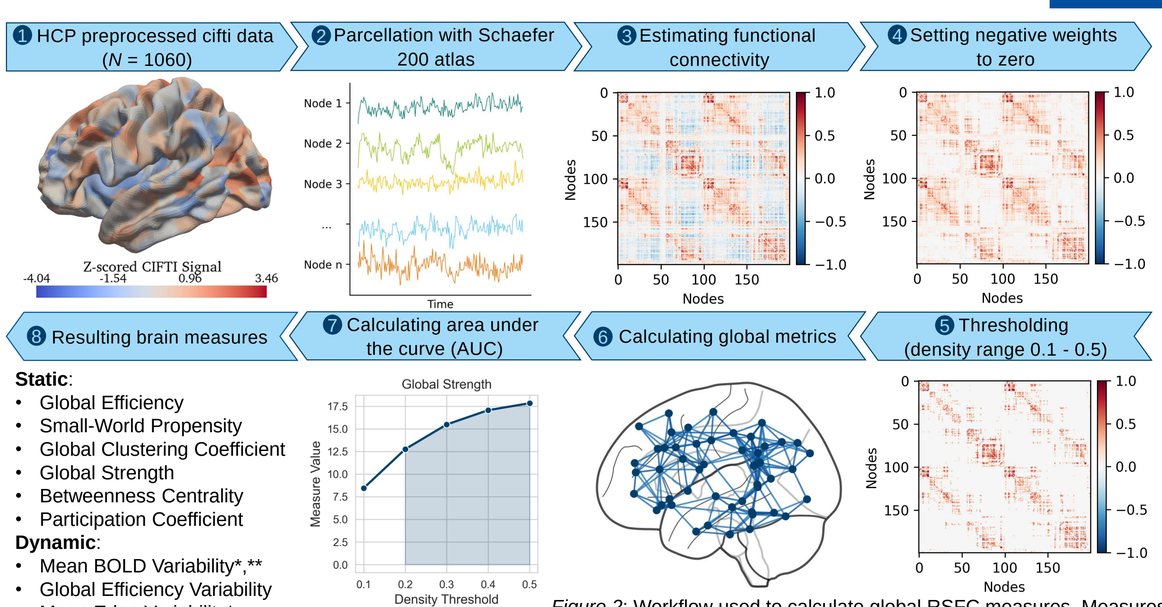
fit	mirt bifactor	lavaan CFA
RMSEA	0.059	0.059
SRMSR	0.051	0.039
TLI	0.966	0.925
CFI	0.973	0.941

Table 1: Fit indices of chosen model

factor loadings		mirt bifactor	lavaan CFA
g	NEORAW_01	0.459	0.390
	NEORAW_06	0.586	0.523
	NEORAW_11	0.705	0.637
	NEORAW_16	0.653	0.552
	NEORAW_21	0.702	0.612
	NEORAW_26	0.794	0.713
	NEORAW_31	0.639	0.541
	NEORAW_36	0.520	0.462
	NEORAW_41	0.752	0.670
	NEORAW_46	0.540	0.412
	NEORAW_51	0.671	0.588
	NEORAW_56	0.620	0.555
s_anx	NEORAW_01	0.465	0.394
	NEORAW_31	0.403	0.434
s_dep	NEORAW_16	0.394	0.432
	NEORAW_46	0.438	0.401

Table 2: Loadings of chosen mirt model and lavaan CFA

Brain Measures



Static:
- Global Efficiency
- Small-World Propensity
- Global Clustering Coefficient
- Global Strength
- Betweenness Centrality
- Participation Coefficient

Dynamic:
- Mean BOLD Variability*,**
- Global Efficiency Variability
- Mean Edge Variability*

Figure 2: Workflow used to calculate global RSFC measures. Measures marked (*) bypassed steps 4, 5, 7; (**) additionally bypassed step 3.

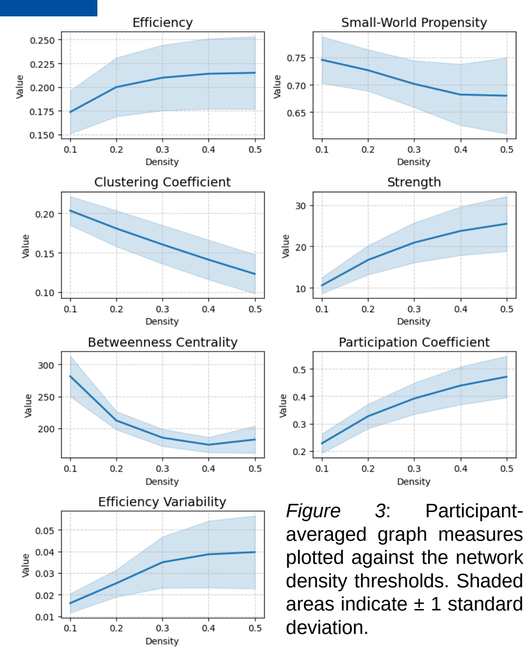


Figure 3: Participant-averaged graph measures plotted against the network density thresholds. Shaded areas indicate ± 1 standard deviation.

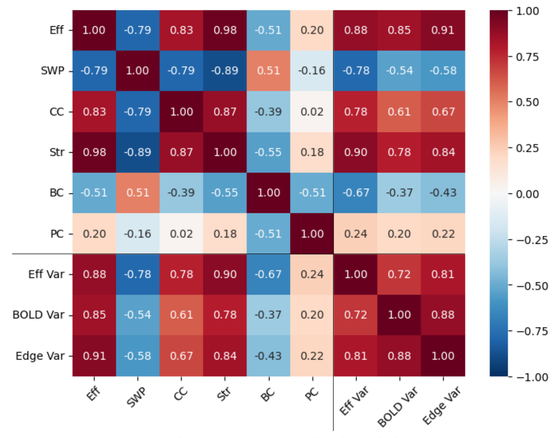


Figure 4: Correlation heatmap of global brain measures. Values represent the AUCs integrated across density thresholds 0.2–0.5. Static measures: Efficiency (Eff), Small-World Propensity (SWP), Clustering Coefficient (CC), Strength (Str), Betweenness Centrality (BC), Participation Coefficient (PC); Dynamic measures: Efficiency Variability (Eff Var), BOLD Variability (BOLD Var), Edge Variability (Edge Var).

Integrated SEM

Integrated Structural Equation Model (SEM):

- **Sample**: *N* = 1053 (intersection of participants providing both neuroticism data and brain imaging data)
- **Measurement Model Component:**
  Implemented the optimal bifactor structure identified in the preceding analysis (see "Measurement Model" section)
- **Structural Model Component:**
  - **Standardization**: All brain measures were z-transformed; for graph metrics, this utilized AUC values integrated across density thresholds (0.2-0.5)
  - **Predictor selection**: A subset of brain measures was selected based on the correlation matrix to reduce multicollinearity

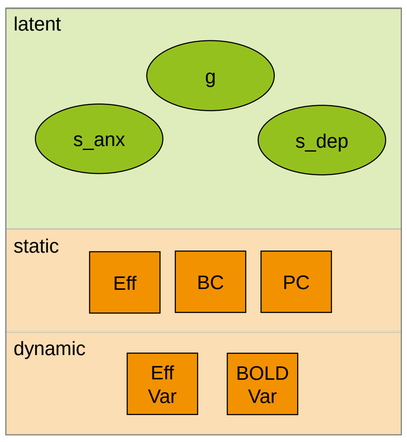


Figure 5: Illustration of included brain measures and latent variables

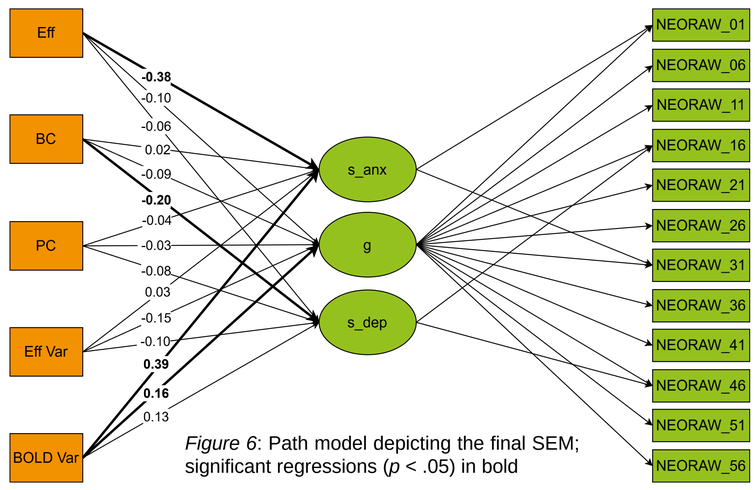


Figure 6: Path model depicting the final SEM; significant regressions (p < .05) in bold

fit	lavaan SEM
RMSEA	0.044
SRMSR	0.031
TLI	0.922
CFI	0.940

Table 3: Fit indices of the integrated SEM

latent factor	explained variance (R^2)
g	1.3 %
s_anx	4.5 %
s_dep	2.1 %

Table 4: Explained variance of the model's latent factors

IV. Discussion

Conclusion
- Excellent bifactor model fit supports a **multidimensional latent structure** for **neuroticism** in the HCP dataset
- Global RSFC markers (including **dynamic BOLD Var, static Eff**, and **BC**) significantly **associate with latent neuroticism**
- However, **collective impact** remains **negligible**, explaining 1-5% of total variance

Outlook / Limitations
- **NEO-PI-R** (48 items): Capturing all facets beyond the 12-item NEO-FFI subscale
- **Multiverse Analysis**: Testing robustness across varied atlases, connectivity methods, thresholds, etc.)
- **State-Based Metrics**: Incorporating e.g. switching rate to capture temporal fluctuations
- **Network-Level FC**: Shifting from global to system-specific measures to prevent masking divergent or opposite effects

Supplementary Materials
Scan QR code to access full documentation for:
- Neuroticism data preparation
- Tested measurement models
- Brain measure calculations
- SEM results

OPEN DATA (PA) OPEN MATERIALS

V. Acknowledgements

Data were provided by the Human Connectome Project, MGH-USC Consortium (Principal Investigators: Bruce R. Rosen, Arthur W. Toga and Van Wedeen; U01MH093765) funded by the NIH Blueprint Initiative for Neuroscience Research grant; the National Institutes of Health grant P41EB015896; and the Instrumentation Grants S10RR023043, 1S10RR023401, 1S10RR019307.
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VI. References

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